

Electronic Design Automation Using Altium Designer 15

JAI SACHITH PAUL

Electronic product design is never a single-step process. Right from conceptualisation of the idea to its fabrication and packaging, the design should account for a variety of factors. The traditional electronic design environments make use of a divide-and-conquer approach. Here, each step is implemented separately on different platforms. These are later integrated to form a single design. This might result in the omission of some minute but crucial details, besides reframing and eating-up your limited budget. Also, you might not be able to meet your project deadlines, which can risk the time to market.

Altium Designer from Altium is a one-stop electronic design environment that integrates schematic capture, PCB design, FPGA design, embedded-software development and manufacturing file support into a single unified application. Ultimately, this can help you eliminate data-translation and synchronisation issues.

What is new in Altium Designer 15

Altium Designer 15, the latest version of the software released in November 2014, comes with a powerful array of new features and enhancements to its core technologies. This version has also addressed some issues that were raised by the customers. Let us take a look at some of them.

xSignals for high-speed designs.

The signal-propagation rate in modern designs is very high and can even rise to 100 gigabytes per second. Deciding the design rules is not easy at these

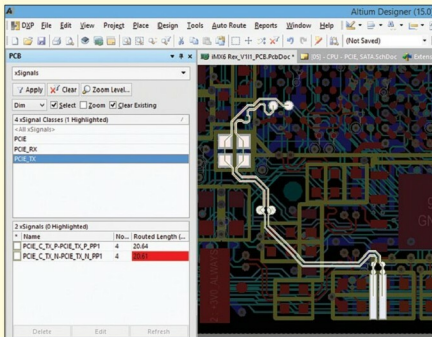


Fig. 1: xSignals in Altium Designer 15

propagation rates. The reason is simple. While the designer sees the function of the signal, PCB editor can only view the signal as a set of connected pins. This makes it difficult for the design engineer to specify key requirements of his design, such as length and matched-length design rules.

The new feature from Altium called xSignals is introduced primarily to resolve this issue. xSignals helps the designer to treat the high-speed signal path in exactly the way he wants. That is, as a path in which the signal travels between the source and destination covering the termination components and branches. This path helps the designer to accurately plan and route groups of high-speed nets with more efficiency.

Solder mask-expansion enhancements. In today's PCB designs,

xSignal

xSignal refers to a signal path between two nodes defined by the designer. This path can either be between two nodes within a net or in associated nets separated by a component. We can use xSignals to scope relevant design rules that are to be obeyed during design tasks, such as interactive length tuning.

there are situations where we need different expansion settings for top and bottom layers of applied solder masks. Altium Designer 15 is now equipped with pad and via-expansion settings by which we can have individual configurations for top and bottom layers.

PCB rules and constraints editor can be used for defining and modifying solder mask-expansion rules for the board pad and vias.